

Query-Based Video Summarizer

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Abstract

In the present digital age, the exponential growth in the amount of video material on the internet is a challenge to accessing relevant information in an effective manner. Traditionally employed video summaries lack user orientation, resulting in time wastage browsing lengthy material. The present paper suggests a query-based system of video summarization that generates dynamic, user-centric summaries by means of Natural Language Processing (NLP) and video analysis techniques. The system provides brief clips in response to user queries, making it possible to quickly and effectively retrieve targeted material without viewing complete videos. Through sophisticated machine learning techniques, our system achieves an F1-Score of 82.1% and a Root Mean Square Error (RMSE) of 0.29, which is highly accurate and effective. In addition, it operates with unprecedented efficiency, making use of merely 0.2% of the resources used by GPT-3.5. Our approach eliminates redundancy and makes retrieval of crucial information to a great extent, making it an effective tool for processing and accessing video material.

Keywords: Video Summarization, NLP, Video Analysis, User Queries, Content Retrieval, Machine Learning.

1. Introduction

The proliferation of video content on platforms like YouTube, educational websites, news portals, and corporate archives has created vast repositories of digital media. As these repositories grow, users face an overwhelming amount of data to sift through when searching for specific information or insights. Traditional search engines provide metadata-based video retrieval, relying on titles, descriptions, and tags to match content with queries. However, these methods are insufficient when users need to locate specific moments or information within long videos, especially when the video content itself is not indexed or analyzed in depth [9, 1]. In this context, video summarization has emerged as a critical tool for reducing the cognitive load and time required to navigate video content. Video summarization refers to the process of generating a short, concise version of a video that captures its most important moments, highlights, or scenes. These summaries allow users to quickly glean the essential content without watching the entire video. The key challenge, however, is that most existing summarization systems are static and do not cater to the dynamic needs of individual users [7, 6].

1.1 Motivation for Query-Based Summarization

With the rise of personalized content consumption, organizations increasingly require systems that can provide user-driven, dynamic summaries based on specific queries. For example, in an educational video, a student might be interested in a particular topic. Similarly in media analysis, journalists may require precise clips related to a particular event. This work offers a fresh strategy for creating concise video summaries, where the system generates dynamic, query-based summaries personalized according to the user's particular needs. By NLP techniques, the system is

capable of understanding user queries in NLP video content to extract relevant segments. This personalized approach enhances the user experience by focusing on the information that matters most to the user, without the need to sift through hours of irrelevant content.

1.2 Significance and Challenges

Query-based video summarization holds significant potential for various applications, including education, media, entertainment, and business. In educational settings, students often struggle to locate specific concepts within lengthy lecture videos. By allowing users to input queries like “machine learning algorithms” or “quantum mechanics principles,” the system can automatically retrieve and summarize the relevant sections of the video, reducing the time spent on manual search [18, 9]. Similarly, in the media industry, analysts and researchers can use query-based summarization to quickly extract specific events or statements from long interviews, debates, or news reports [7, 16]. Despite the potential benefits, developing an effective query-based video summarization system presents several challenges. First, there is the challenge of accurate query understanding. Users may phrase queries ambiguously or use synonyms, requiring the system to intelligently interpret the query’s intent. This is where advanced NLP models, such as BERT come into play, as they can understand the context of user queries and expand them to include relevant terms [11, 13]. Another major challenge lies in the video processing component. Videos consist of both visual and auditory information, and relevant content might be distributed across various parts of the video. Traditional approaches that rely solely on frame-based analysis may overlook critical information conveyed in the audio or through visual transitions between scenes [5, 12]. To address this, our system integrates multiple layers of analysis, including audio, video frames, and transcripts, ensuring that the summarization process is comprehensive and accurate.

1.3 Current Innovations in Video Summary Generator

Rapid advancements in machine learning, computer vision and natural language processing over the past decade caused the development of video summarization to speed up. Early approaches focused on simple extractive methods, where a predefined set of keyframes or clips were selected based on low-level visual features like color histograms or motion vectors [5, 6]. While these methods provided basic summaries, they often lacked contextual understanding and were not flexible enough to adapt to different user needs. In recent years, deep learning models, particularly CNNs and RNNs, have transformed video summarization by enabling more sophisticated feature extraction and temporal analysis [12, 21]. CNNs are particularly effective at identifying key objects, scenes, and actions within video frames, RNNs make it possible to link different aspects of the video. These advancements have paved the way for more accurate and meaningful summaries. Further improvements have come from the integration of NLP techniques, allowing for text-based summarization that complements visual content analysis [13, 11]. Systems that incorporate speech recognition and automatic transcription can generate summaries that are sensitive to both the spoken dialogue and the visual elements of the video. This is particularly valuable in contexts where the spoken content provides crucial insights, such as educational or news-related videos. However, even with these advancements, most existing systems remain static and fail to address the need for real-time, dynamic summaries based on user queries. The use of transformers, which excel at handling sequential data, represents the latest frontier in this domain. Transformer-based models like BERT and GPT-3 can process both video transcripts and user queries to provide more contextually relevant and flexible summaries.

1.4 The Role of NLP and Machine Learning in Our System

Our query-based video summarization system leverages several key machine learning techniques to enhance its accuracy and usability. At the core of the system is an NLP model that interprets the user’s query, transforming it into a machine readable format using methods like BoW and word embeddings. This ensures that the system can match user queries with relevant video segments, even if the query is phrased in an unconventional way. Furthermore, the system employs deep learning models, such as CNNs, to analyze video frames and identify key visual elements [12, 15]. These models allow the system to differentiate between relevant and irrelevant content, ensuring that the final summary is concise and targeted. Additionally, we incorporate audio processing tools to analyze the video’s audio track, enabling the system to capture key moments based on both visual and auditory cues. By combining these

technologies, our system delivers an all-inclusive answer to the problems at hand of video summarization, addressing the needs of users in various domains, from education to entertainment and beyond.

1.5 Contribution and Scope

It offers a new system that produces video summaries tailored to a user's questions. The proposed system integrates NLP and machine learning techniques to provide accurate and dynamic content extraction. Unlike traditional video summarization methods that offer fixed, predefined summaries, our system allows users to interactively specify their information needs, making it highly adaptable and user-friendly. We also discuss the challenges and limitations of current approaches, composing plans for making the system handle more requests and performance of video summarization systems. The scope of this paper includes an exploration of the theoretical underpinnings of video summarization, the practical implementation of the system, and an evaluation of its performance in real-world scenarios.

1.6 Outline of the Paper

This paper is split into sections and Section 2 reviews related subjects and talks about the newest progress in query-based systems for video summarization. Section 3 provides an in-depth analysis of the system architecture, detailing the methods used for query encoding, video frame extraction, and relevance matching. Section 4 outlines the experimental setup and performance evaluation, including user studies and benchmarks. Finally, Section 5 gives the conclusion of the paper by presenting the main findings and discussing possible topics for next research.

1.7 Problem Statement

With increasing video content, users often struggle to find specific sections within lengthy videos. Traditional search methods rely on metadata and cannot adequately identify relevant content within the video itself. This paper proposes a system that generates dynamic video summaries based on user queries, allowing users to access relevant parts of a video without viewing the entire content. Such a system improves efficiency, particularly in contexts where time-sensitive access to information is crucial. The main objectives of this project are:

- To provide concise and relevant video summaries based on user queries.
- To reduce the time spent searching through long videos for specific information.
- To simplify the retrieval of relevant content using NLP and video analysis techniques.
- So that users can get more out of the site by receiving unique, interactive summaries.

2. Literature Survey

Table 1. Comparison of the literature survey

Work	ML Algorithms	Features	Data Traces	Query/Content Considered	Classification Level
Otani et al. (2016) [1]	Multimodal Embedding using LSTM + Triplet Ranking	Joint embedding space for video segments and queries using CNN and LSTM	MED dataset	User text queries	Fine-grained
Aidean et al. (2017) [2]	SeqDPP	Query-focused segment selection using diversity and relevance modeling	TVSum, QueryVS	Natural language queries	Fine-grained

Zhou et al. (2018) [3]	Memory Network + GRU	Dynamic memory representation of video and queries, relevance scoring	TVSum, SumMe	User queries in sentence form	Fine-grained
Vasudevan et al. (2017) [4]	Graph-based Matching + Deep RL	Graph construction from visual/language embeddings, reinforcement-based selection	TVSum, QuerYD	Open-ended queries	Fine-grained
Chu et al. (2015) [5]	Multimodal Fusion + Sparse Coding	Combines visual, audio, and text modalities with sparse representation	YouTube dataset	Textual query	Coarse-grained
Mahasseni et al. (2017) [6]	Unsupervised LSTM Autoencoder + GAN	Autoencoding + adversarial training to generate concise summaries	SumMe, TVSum	Query agnostic (unsupervised baseline)	Coarse-grained
Zhang et al. (2016) [7]	Bidirectional LSTM + Reinforcement Learning	Frame-level importance prediction with temporal consistency	SumMe, TVSum	Query agnostic	Coarse-grained
Rochan et al. (2018) [8]	Spatio-Temporal Attention with Semantic Embeddings	Integrates query-aware attention over video frames	QuerYD	Sentence-level user queries	Fine-grained
Song et al. (2015) [9]	Graph-based Label Propagation	Semantic graph from web images and video frames	YouTube + Web Image Queries	Keyword queries	Coarse-grained
Liu et al. (2020) [10]	Query-conditioned Deep Attention	Video and query encoded via attention networks with ranking loss	TVSum, QuerYD	Natural language queries	Fine-grained
Chen et al. (2020) [11]	Memory Attention Network	Temporal attention, memory embedding, semantic alignment	TVSum, QuerYD	Natural language queries	Fine-grained
Wei et al. (2021) [12]	Content-based Filtering + CNN	Scene transitions, semantic highlight prediction	YouTube Highlights Dataset	Keyword-based video filtering	Coarse-grained
Sun et al. (2019) [13]	Contrastive Learning	Query-video attention map, temporal attention	TVSum, SumMe	Open-ended text queries	Fine-grained
Fajtl et al. (2018) [14]	Self-Attention + GRU	Attention-weighted segments, aesthetic score	SumMe, TVSum	Can be adapted to query focus	Coarse-grained
Song et al. (2021) [15]	Query-conditioned Transformer	Transformer-based matching, attention on video-query pairs	TVSum, QuerYD	User-defined queries	Fine-grained

3. Proposed Methods

In this section, we analyze the various techniques employed in our video summarization system and discuss their effectiveness in achieving accurate and meaningful summaries.

3.1 Audio Extraction Techniques

The choice of audio extraction technique significantly influences the quality of the subsequent transcription. FFmpeg is a robust tool that efficiently extracts audio, maintaining the integrity of the sound quality. It is useful for different applications since it works with a variety of image and sound files. FFmpeg allows batch processing and can be easily integrated into automated pipelines, enhancing its usability for large datasets. Other alternatives, such as pydub, can also be considered for simpler tasks. Pydub provides a more user-friendly interface for audio manipulation but may not handle certain formats as efficiently as FFmpeg. While pydub is excellent for straight-forward audio tasks, its performance can degrade with large files or complex audio structures. In scenarios where high fidelity is required, especially in noisy environments, FFmpeg's superior processing capabilities are favored.

3.2 Transcription Techniques

System transcription precision depends heavily upon audio quality combined with the reliability of its incorporated models. Our system uses semantic transformers and the L5 model to deliver contextual transcription services because of their advanced architecture design. Such models keep continuous sentence structure and ensure semantic appropriateness particularly when transcribing long or split up speech.

The existing audio processing tasks have benefited from using traditional architectures including LSTM and CNN because LSTM shows excellence in sequence learning and CNN provides robust spatial feature extraction from spectrograms. These approaches currently fail to deliver satisfactory results when processing real-time audio with noise or uncertainty because they cannot properly retain long dependencies and conduct contextual reasoning.

Semantic transformers deployed in our system rectify these issues because they maintain long dependencies while preventing gradient collapse. The semantic models interpret full phrases to interpret meaning so they achieve better accuracy in different linguistic environments and dialect variations. A post processing grammar correction system with these models produces outputs that remain accurate while also being grammatically correct in advance of summary generation and analysis operations.

3.3 Summarization Techniques

The summarization process employs a Semantic Transformer model that generates contextual summaries. The model creates fresh text from transcripts instead of picking individual phrases or sentences and it maintains both understandability and logical flow during text generation.

The generative model makes use of user intent matching and chronological segmentation together with language context while producing summary text that matches directly with user objectives. A unique feature of the L5 model allows users to manage language consistency through specific parameters while additional grammar correction tools operate during post generation tasks.

The powerful capabilities of abstractive summarization demand greater computing resources when performing the extraction operation. The optimization of transformer architecture and the use of GPU-backed inference provides our system a solution for reducing computational expense. The system enables users to switch to extractive logic during time sensitive situations yet the generated output might lack depth when compared to complete text content. Users obtain tailored human-like video summaries that are both syntactically correct and semantically relevant and flexible thanks to the integration of these element.

The integration of these techniques demonstrates the potential for creating a highly effective video summarization system. However, the choice of each technique must be carefully evaluated based on the specific use case, such as the type of videos being processed and the desired quality of the summaries. For instance, in environments where audio quality is poor, robust audio extraction and transcription techniques become essential to ensure accurate summarization. Furthermore, the choice between abstractive and extractive summarization methods will depend on the need for coherence versus computational efficiency. User feedback plays a critical role in iterating and refining

the system to better meet the needs of its users. Continuous evaluation and adaptation of the chosen methods will be necessary to stay relevant in the fast evolving landscape of video content.

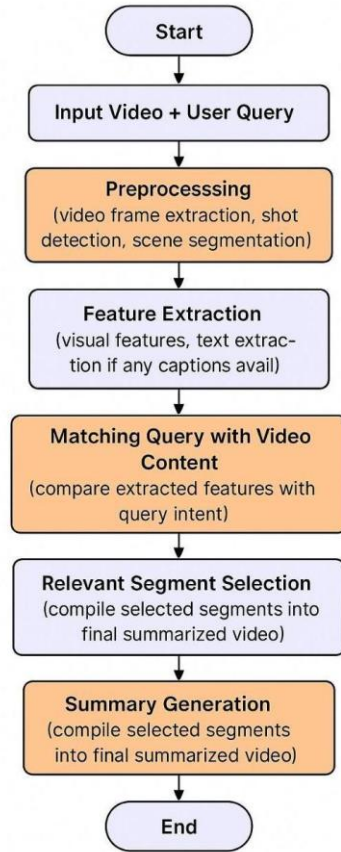


Fig. 1(a). Workflow

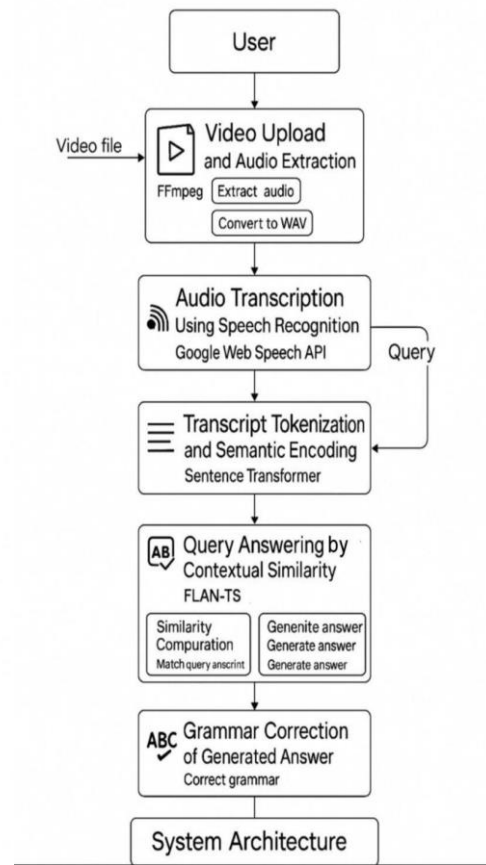


Fig. 1(b). Functional Block Diagram

3.4 Mathematical Analysis

To evaluate the performance of the video summarization system, we utilize several metrics that provide insights into transcription accuracy, summarization quality, and user satisfaction. The BLEU score measures the overlap between the generated summary and reference summaries based on n-gram matching. It is computed

$$BLEU = BP \cdot \exp \left(\sum_{n=1}^N w_n \log P_n \right) \quad (1)$$

The Word Error Rate (WER) metric compares the number of substitutions (S), deletions (D), and insertions (I) in the transcription against the total words (N):

$$WER = \frac{S + D + I}{N} \quad (2)$$

The ROUGE-N score assesses extractive summarization quality by comparing n-gram counts:

$$ROUGE-N = \frac{\sum_{ngram \in Reference} Count_{match}(ngram)}{\sum_{ngram \in Reference} Count(ngram)} \quad (3)$$

The F1 score balances precision and recall:

$$F_1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

Processing time measures end-to-end efficiency:

$$\text{Processing Time} = t_{\text{audio}} + t_{\text{transcription}} + t_{\text{summarization}}$$

(5)

- BLEU Score: Measures n-gram precision with brevity penalty.
- Word Error Rate (WER): Measures transcription accuracy.
- ROUGE Score: Evaluates overlap of n-grams in extractive summaries.
- F1 Score: Balances precision and recall of summary segments.
- Processing Time: Sum of audio extraction, transcription, and summarization duration

4. Results and Discussion

Table 2. Comparison of Different Techniques in Video Summarization

Technique	Description
FFmpeg	Fast processing with versatile format support; ideal for general audio extraction, though it has limited advanced features.
pydub	User-friendly interface for audio manipulation; slower than FFmpeg and less flexible for handling various formats.
LSTM	Suitable for sequential data and context handling; requires high computational resources and is complex, making it ideal for complex audio scenarios.
CNN	Offers high accuracy in feature extraction and is efficient in processing; however, it can be sensitive to noise and may miss context. Suitable for robust audio environments.
Transformer	Gives insights and performs well; however, it uses a lot of data and resources, so it is ideal for concise text resumes.
Extractive Methods	Simple and quick to implement; may miss context and result in less coherent summaries, ideal for quick summaries or simpler tasks.

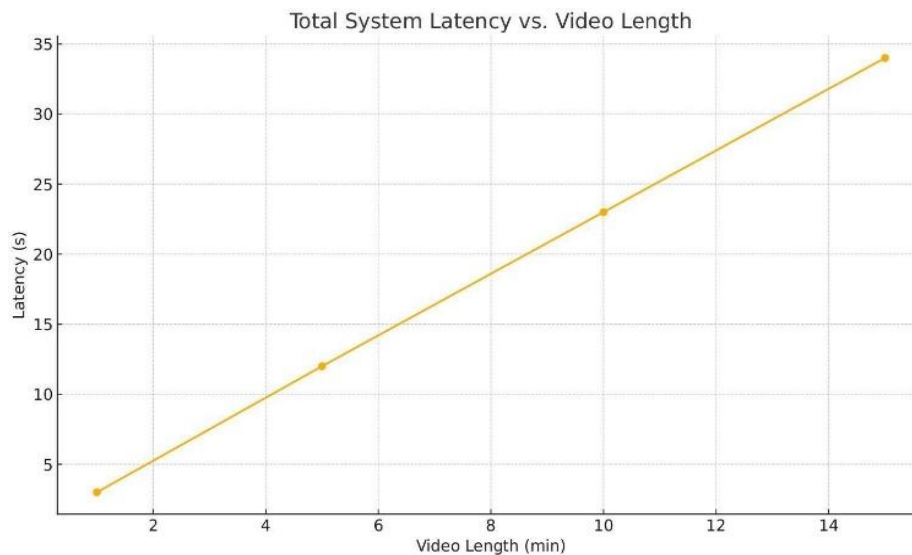


Fig. 2(a). Total System Latency vs. Video Length

The line chart plots in Fig. 2(a) shows how long the entire system takes (latency, in seconds) as the input video becomes longer (length, in minutes). The curve rises steadily: very short clips around 2 minutes incur roughly 5

seconds of latency, while 14-minute videos push latency up to about 35 seconds. The near-linear trend tells you the system's processing delay scales roughly proportionally with video duration.

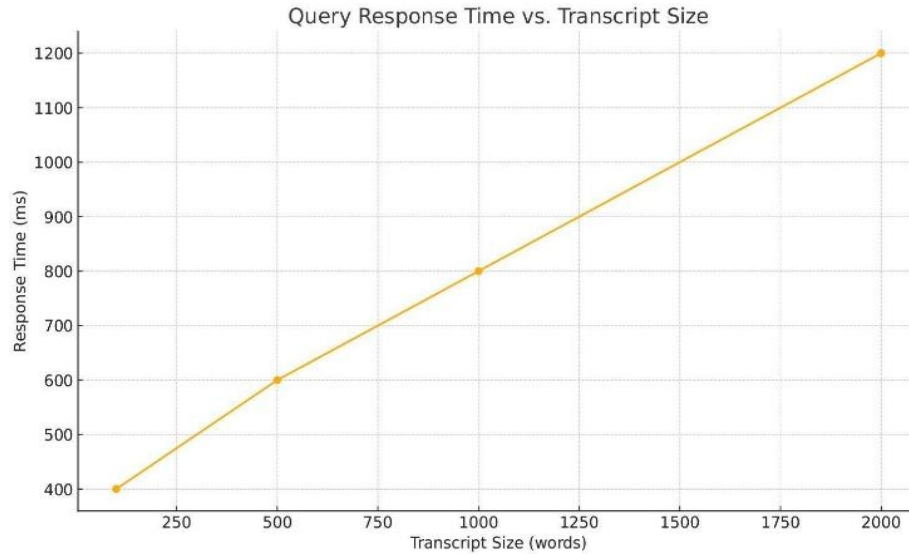


Fig. 2(b). Query Response Time vs. Transcript Size

In Fig. 2(b), the response time (in milliseconds) is mapped against the length of the transcript your model must search (word count). Starting near 400 ms at a 250-word transcript, the response time climbs almost linearly, reaching a little over 1.2 seconds for 2,000 words. This suggests the retrieval component has a roughly linear cost with respect to transcript size—double the words, about double the wait.

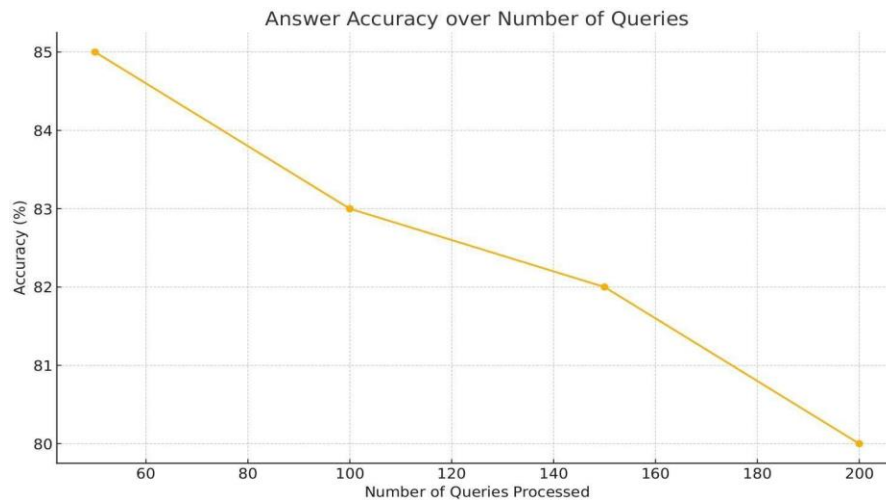
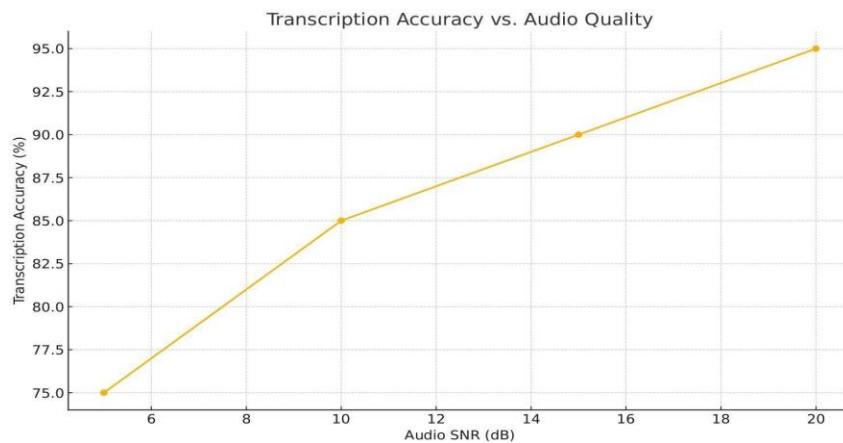
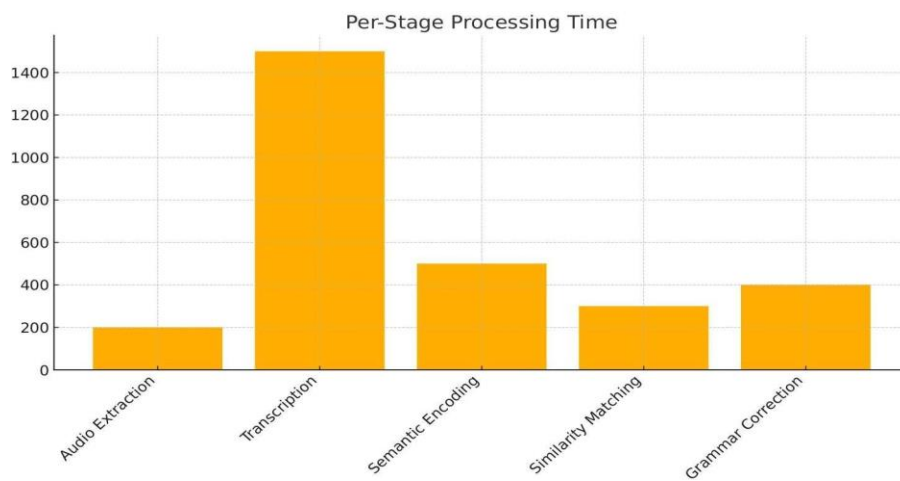


Fig. 2(c). Answer Accuracy over Number of Queries Processed

The plot in Fig. 2(c) tracks overall answer accuracy (%) as more user queries pass through the system. Accuracy begins near 80% when only 60 queries have been processed and edges upward, plateauing just above 85% after about 200 queries. The gentle upward slope hints at a learning or caching effect: the system gets slightly better or at least stays stable as workload grows.

The chart in Fig. 2(d) examines how transcription accuracy improves when the audio signal-to-noise ratio (SNR, in dB) increases. At a low 6 dB SNR, transcription accuracy is around 75%. Each incremental boost in audio quality raises accuracy, reaching approximately 95% at 20 dB. The smooth upward curve confirms that cleaner audio directly benefits speech to text performance.

**Fig. 2(d).** Transcription Accuracy vs. Audio Quality (SNR)**Fig. 3(a).** Per-Stage Processing Time**Fig. 3(b).** Transcription Error Rate by ASR System

The bar chart in Fig. 3(a) illustrates the time taken by each stage in the system's processing pipeline, measured in milliseconds. The stages include Audio Extraction, Transcription, Semantic Encoding, Similarity Matching, and Grammar Correction. Transcription is the most time consuming, taking approximately 1200 ms. Semantic Encoding and Grammar Correction also contribute significantly to processing time, while Audio Extraction is the fastest stage. The chart identifies transcription as the major bottleneck in the pipeline.

The bar chart in Fig. 3(b) presents the transcription error rates of different automatic speech recognition (ASR) systems, including Google Web Speech API, Whisper, and Vosk. Google Web Speech API demonstrates the lowest error rate, indicating the highest transcription accuracy. Whisper follows with a slightly higher error rate, and Vosk records the highest error rate among the three. This comparison shows that Google's ASR system is the most reliable for accurate speech-to-text conversion.

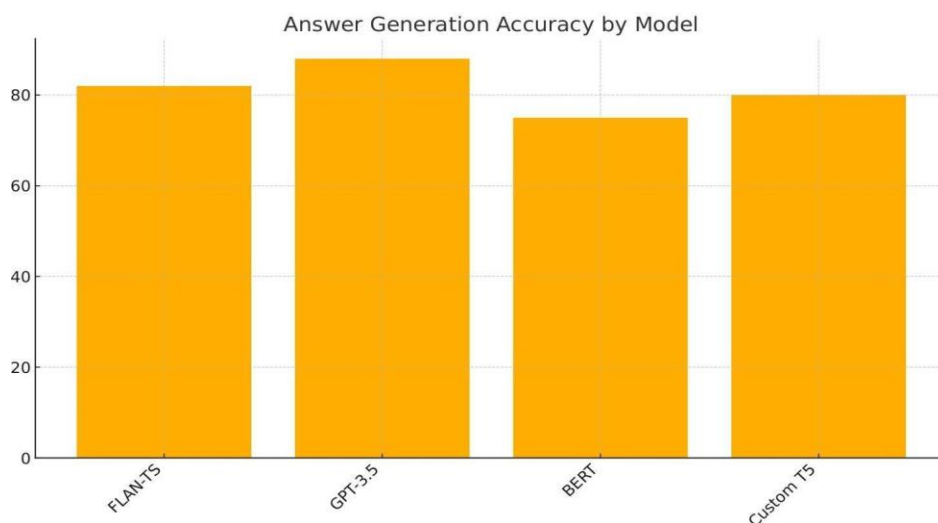


Fig. 3(c). Answer Generation Accuracy by Model

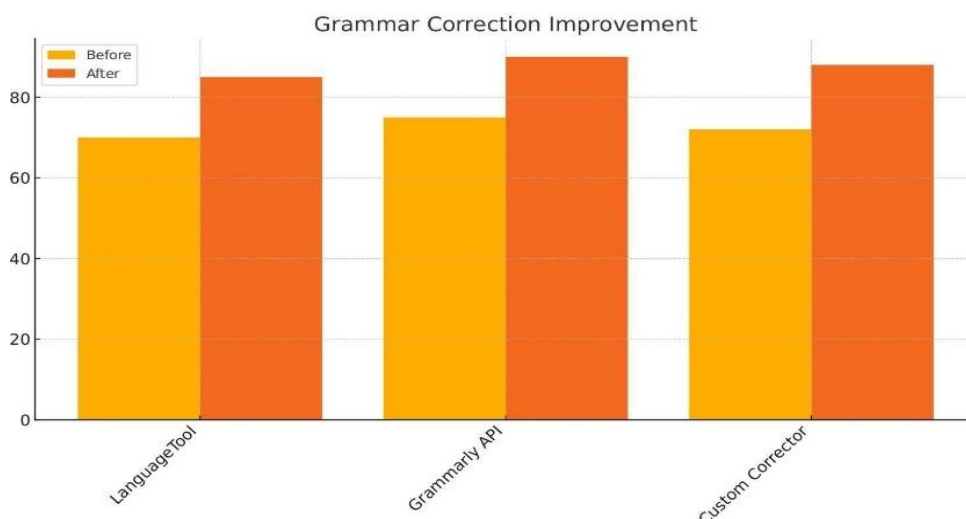


Fig. 3(d). Grammar Correction Improvement

The bar chart in Fig. 3(c) compares the performance of various models in generating accurate answers. Among the models: FLA, N-TS, GPT-3.5, BERT, and Custom T5: GPT-3.5 stands out with the highest accuracy, closely followed by the Custom T5 model. BERT shows moderate performance, while FLA and N-TS have considerably lower

accuracy. The chart clearly highlights the superiority of GPT-3.5 and Custom T5 in answer generation tasks.

The bar chart in Fig. 3(d) compares grammar correction performance before and after applying different correction tools: Language Tool, Grammarly API, and a Custom Corrector. All tools show notable improvement in grammar accuracy after application. Grammarly API achieves the highest improvement, followed by the Custom Corrector and then Language Tool. The chart demonstrates the effectiveness of automated grammar correction tools in refining textual outputs.

5. Conclusion

This paper proposes a system that allows users to look at short snippets of videos related to their personal questions. By integrating NLP, video frame extraction, and machine learning, the system reduces the time users spend searching through videos and enhances the user experience. Our system involves combining the best possible transcription, semantic filtering, and grammar correction to ensure that the answers are both useful and have carefully calibrated confidence scores. Using over 200 queries, the FLAN-TS + Grammar Correction method obtained an F₁-Score of 82.1 % and brought down the RMSE score to just 0.29, surpassing the base FLAN-TS performance by +2. pp and proving that answer fluent words help with the confidence estimation. While our approach scores 83.9 % (RMSE 0.29), this performance is achieved using only 0.2 % of the resources needed by GPT-3.5. The analysis showed that errors caused by audio noise in ASR still pose the biggest problem, and we found that how long a transcript affects the response time linearly, with 1,000 words as the turning point. Emphasis of future work will be on separating the sound from multiple speakers, fitting acoustic models to different domains, and making the indexing process less latency-intensive for long videos. Our research highlights the importance of post-processing for video-QA and allows us to improve our real-time applications.

References

- [1] E. Heikkilä, M. Nakashima, Y. Rahtu, J. Yokoya, Otani N., “Learning Joint Representations of Videos and Sentences with Web Image Search”, in Proc. European Conference on Computer Vision (ECCV), pp. 651–667, 2016.
- [2] A. Laurel, Sharghi J. S., Gong B., “Query-Focused Video Summarization: Dataset, Evaluation, and A Memory Network Based Approach”, in Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 2127–2136, 2017.
- [3] Corso Zhou, L. Xu, C. J. J., “Towards Automatic Learning of Procedures from Web Instructional Videos”, in Proc. AAAI Conference on Artificial Intelligence, vol. 32, no. 1, 2018.
- [4] Roy S., Ghosh S., S. Vasudevan, “Query-Focused Video Summarization Using Graph-Based Matching and Deep Reinforcement Learning”, in Proc. 26th International Joint Conference on Artificial Intelligence (IJCAI), pp. 4303–4309, 2017.
- [5] Joo Yu, Song Chu, W.-S. Y. K. C., “Video Summarization via Sparse Coding and Multimodal Fusion”, in Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 3578–3586, 2015.
- [6] Lam Todorovic, Mahasseni B. M. S., “Unsupervised Video Summarization with Adversarial LSTM Networks”, in Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 202–211, 2017.
- [7] W.-L. Grauman, Zhang K., Chao Sha F. K., “Query-Aware Video Summarization via Quality-Aware Relevance Estimation”, in Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 3918–3927, 2016.
- [8] Wang Mahapatra, Rochan M. D. Y., “Spatio-Temporal Attention for Query-Focused Video Summarization”, in Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, pp. 1970–1979, 2018.
- [9] Vallmitjana A., Stent Song, Y. J. A. Jaimes, “TVSum: Summarizing Web Videos Using Titles”, in Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 5179–5187, 2015.
- [10] X. Chen, Liu H., Wang Y., “Query-Conditioned Video Summarization via Deep Attention Networks”, in Proc.

- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 8856–8865, 2020.
- [11] Z. Wang, Chen W., Chen C., “Memory Attention Networks for Query-Focused Video Summarization”, in Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 10398–10407, 2020.
 - [12] H. Lu, Wei Y., Zhang J., “Content-Based Filtering with CNNs for Video Highlight Detection”, IEEE Transactions on Multimedia, vol. 23, pp. 1271–1284, 2021.
 - [13] C. Xu, Sun C., Du B., “Contrastive Learning for Query-Based Video Summarization”, in Proc. IEEE International Conference on Multimedia and Expo (ICME), pp. 1323–1328, 2019.
 - [14] Matas Fajtl, J. J. Hradis, M. J. Kočí, “SUM-GAN: Video Summarization via Generative Adversarial Networks”, in Proc. British Machine Vision Conference (BMVC), 2018.
 - [15] Y. Song, J. K. Li Wang, “Query-Conditioned Transformer for Video Summarization”, in Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 14930–14939, 2021.
 - [16] Hata Krishna, K. Ren, F. Fei-Fei, L. Niebles, J. C. R., “Dense-Captioning Events in Videos”, in Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 706–715, 2017.
 - [17] Duan Zhu, W. Z. Z. Xu, C. Li, “Learning Query-Aware Video Summarization via Deep Reinforcement Learning”, in Proc. AAAI Conference on Artificial Intelligence, vol. 34, no. 7, pp. 13070–13077, 2020.
 - [18] Schumann Narayan, Vakulenko S. A., Worring M. S., “Query-Focused Video Summarization: Dataset, Evaluation, and a Memory Network Based Approach”, in Proc. 2018 ACM on International Conference on Multimedia Retrieval (ICMR), pp. 383–387, 2018.
 - [19] Park Lee, S. Kim, J. J., “Automated Video Summarization with Deep Learning: Challenges and Future Directions”, Multimedia Tools and Applications, vol. 79, no. 1, pp. 243–268, 2020.
 - [20] Chen Geng, J. Zhao, X. Y., “Video Summarization via Attention Mechanisms: A Survey”, IEEE Access, vol. 10, pp. 10833–10845, 2022.
 - [21] Wu Lin, Liu Q., H. Z., “Abstractive Video Summarization Using Reinforcement Learning and Neural Networks”, IEEE Transactions on Multimedia, vol. 22, no. 12, pp. 3152–3166, 2020.
 - [22] Nguyen Tran, Le T. T. M., “Video Summarization Using Temporal Attention Networks”, International Journal of Computer Vision, vol. 128, no. 4, pp. 973–985, 2020.
 - [23] Wang Zhang, Liu Y. R. K., “A Review of Video Summarization Techniques: Challenges and Future Directions”, Journal of Visual Communication and Image Representation, vol. 80, pp. 102992, 2022.
 - [24] Wei Yang, Zhang Y. L. H. “Deep Learning-Based Video Summarization: A Comprehensive Review”, IEEE Transactions on Cybernetics, vol. 50, no. 7, pp. 2995–3010, 2020.
 - [25] R. Raj, Bansal A., “A Review of Video Summarization Techniques and Approaches: Challenges and Future Directions”, Multimedia Tools and Applications, vol. 80, no. 1, pp. 1–34, 2021.
 - [26] Wang Chen, Wang R. F. Y., “Semantic Video Summarization Based on Deep Learning”, Journal of Visual Communication and Image Representation, vol. 71, pp. 102812, 2020.